

Mean or median? Choose correctly.

The wrong average does not just look different — it tells your reader something untrue.

The mean and the median both claim to show 'the typical value'. When data are symmetric they agree, and it does not matter which you use. When data are **skewed** they disagree, and only one of them is honest.

The rule

Situation	Use	Report it as
Data roughly symmetric — mean and median are close	Mean	Mean (SD) — e.g. 48.5 kg (SD 9.1)
Data skewed , or there are extreme values	Median	Median (IQR) — e.g. 7.0 km (IQR 4.2–9.8)
Ordinal variable	Median	Median, or just counts and %
Nominal variable	Neither	n (%) — e.g. 104 (57.8%)

How to tell if your data are skewed — three quick checks

- 1. Compare the mean and the median.** If they are far apart, the data are skewed. This is the fastest check and it costs you nothing.
- 2. Draw a histogram.** A long tail stretching to the right means right-skewed. Ten seconds in Excel, and it will tell you more than any statistic.
- 3. Look at the minimum and the maximum.** A maximum far above the third quartile means a long tail — or a data entry error.

A worked example from the practice dataset

The column *distance_to_facility_km* — how far each patient lives from their treatment facility:

Statistic	Value	What it tells you
Mean	12.4 km	Pulled upwards by a few patients in very remote sucos
Median	7.0 km	The true midpoint: half of patients live closer than this
IQR (Q1–Q3)	4.2 – 9.8 km	The middle half of patients live within this range
Minimum	0.2 km	
Maximum	77.9 km	The long tail that drags the mean up

Why this matters for your programme

If you report the mean, you tell the Ministry that a typical patient travels **12.4 km**. But **three-quarters of patients travel less than 9.8 km**. The mean describes almost nobody. Plan your outreach on the median and the IQR, and describe the long tail separately — because those remote patients are exactly the ones who are lost to follow-up.

How to get these numbers

What you want	In Excel	In R
Mean	=AVERAGE(D2:D181)	mean(df\$age_years, na.rm = TRUE)

What you want	In Excel	In R
Median	=MEDIAN(D2:D181)	median(df\$age_years, na.rm = TRUE)
Standard deviation	=STDEV.S(D2:D181)	sd(df\$age_years, na.rm = TRUE)
Q1 and Q3	=QUARTILE.INC(D2:D181,1) =QUARTILE.INC(D2:D181,3)	quantile(df\$age_years, c(.25,.75), na.rm = TRUE)
Everything at once	Data > Data Analysis > Descriptive Statistics	summary(df\$age_years)

WATCH OUT

AVERAGE() ignores blank cells but not text. If one cell in a number column contains '12 km' instead of 12, Excel will silently treat that row as empty and your denominator will be wrong — with no error message. Always check that COUNT() returns the number of records you expect.

Liafuan importante • Key words in Tetum

Mean, average	Média
Median (middle value)	Valór klaran
Range	Intervalu
Typical, usual	Baibain
High / low	Aas / kraik
Distance	Distánsia